

PythonPowerFlow — Interactive OPF Demo

WB5 Five-Bus Case · All Solvers · All Metrics

<https://github.com/DavidLikesLearning/PythonPowerFlow/tree/DavidProject>

PythonPowerFlow compares five OPF solvers on standard test networks using an interactive CLI (**my_opfs.py**). The WB5 network (Bukhsh et al. 2013) is the canonical example of a non-exact SOCP relaxation: the cheap generator at Bus05 (\$1/MWh) is only reachable via two high-impedance lines, creating a 41% duality gap between the SOCP lower bound (\$740/h) and the true AC optimum (\$1,256/h).

Solvers run: Project SOCP · Project DC OPF · Circuits NR · Panda DC OPF · Panda AC OPF

Bus Voltage Magnitudes

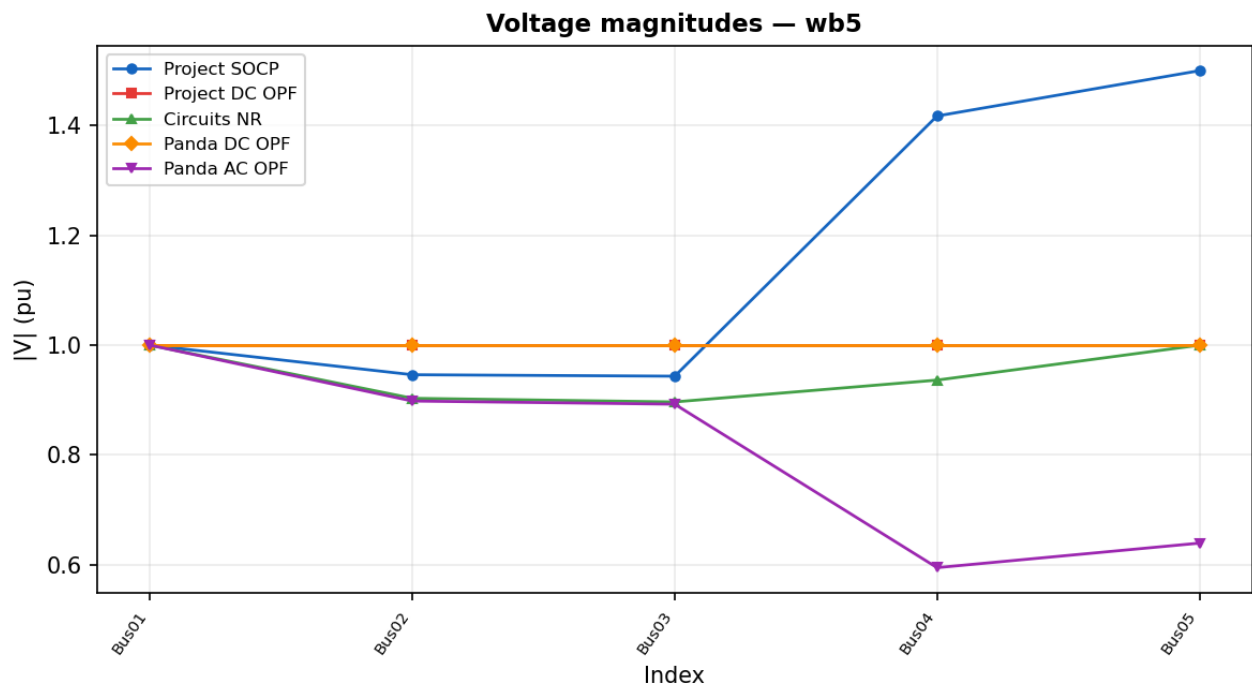


Figure 1. Bus voltage magnitudes (pu). SOCP pushes Bus04–05 to the 1.5 pu upper bound; Panda AC OPF finds the physical solution (Bus04 = 0.59 pu, Bus05 = 0.64 pu); DC solvers fix all voltages at 1.0 pu.

Bus Voltage Angles

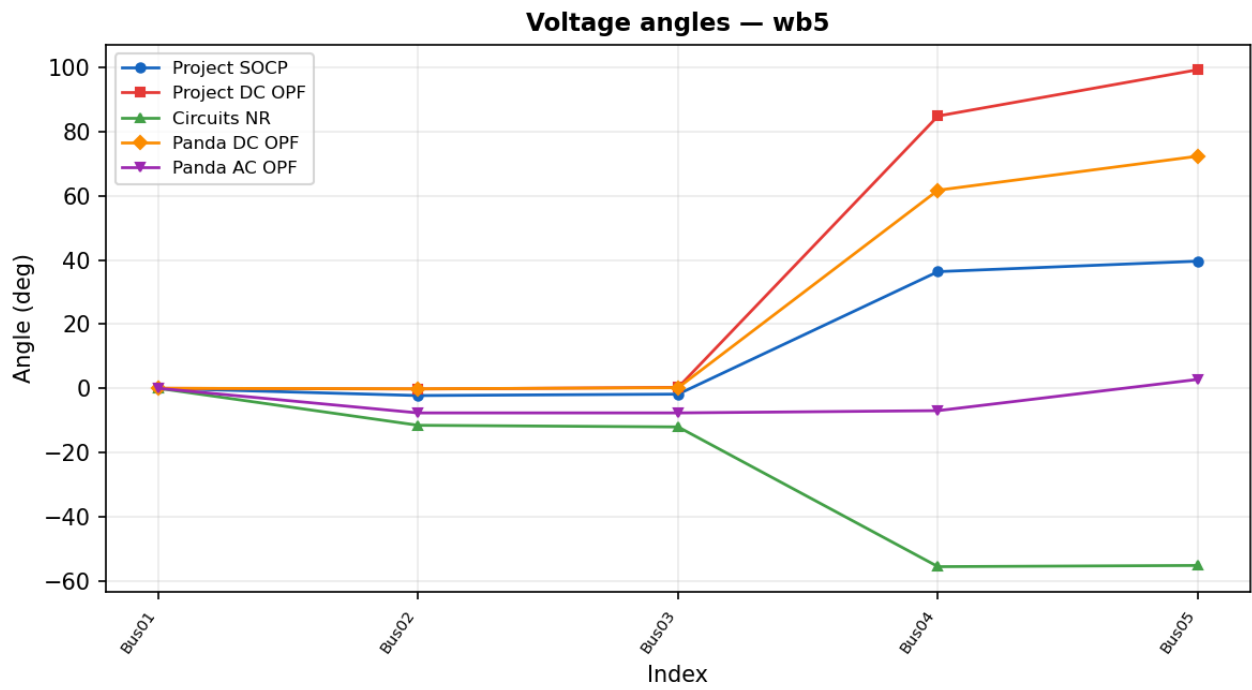


Figure 2. Voltage angles ($^{\circ}$). SOCP and DC OPF show unrealistically large spreads (Bus05 up to $+99^{\circ}$) reflecting relaxed voltages. NR and AC OPF produce physically consistent negative angles.

Duality Gap & SOCP Tightness

Branch tightness $\tau = |W_{ij}|^2 / (W_{ii} \cdot W_{jj})$ measures SOC slack per branch. $\tau = 1$ everywhere is necessary but *not sufficient* for exactness on meshed networks — loop residuals must also be zero.

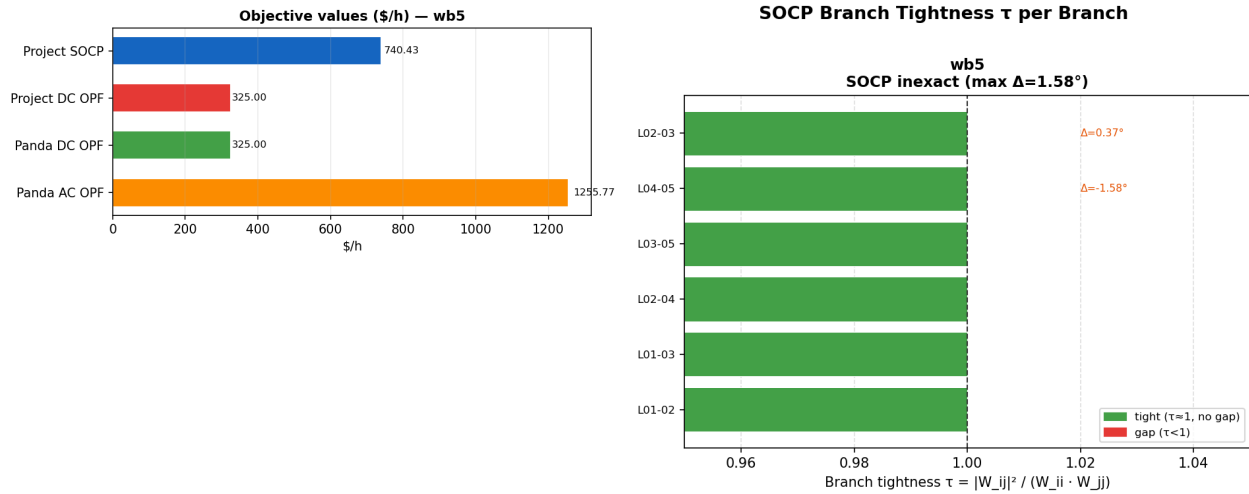


Figure 3 (left). Objective values (\$/h): SOCP (\$740) and DC OPF (\$325) are lower bounds; Panda AC OPF (\$1,256) is the true AC optimum — 41% duality gap. Figure 4 (right). Per-branch SOC tightness τ : all 6 branches reach $\tau = 1$ (gap $< 3 \times 10^{-10}$) — the relaxation is tight everywhere.

WB5 key result — $\tau = 1$ everywhere, yet 41% duality gap:

All 6 SOC constraints are active, but loop residuals (L04-05: -1.58° , L02-03: $+0.37^\circ$) reveal phase inconsistency around the two fundamental cycles. The W matrix is not globally rank-1 — the SOCP optimum is AC-infeasible. This is the canonical illustration that $\tau = 1$ is **necessary but not sufficient** for SOCP exactness on meshed networks.

Branch Active Power Flows

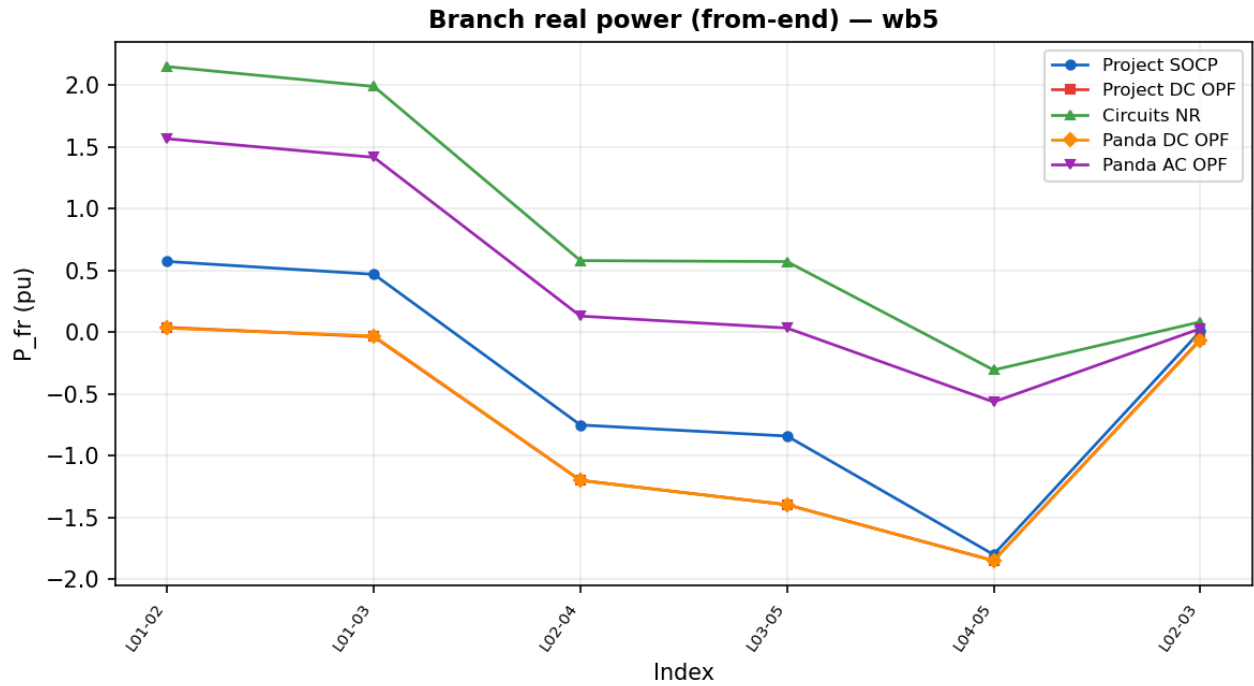


Figure 5. Active power P_{from} (pu). SOCP and DC OPF show negative flows on L02-04 and L03-05 (cheap generator pulling power backwards through high-impedance lines). NR and AC OPF show positive flows at the physical operating point.

Branch Reactive Power Flows

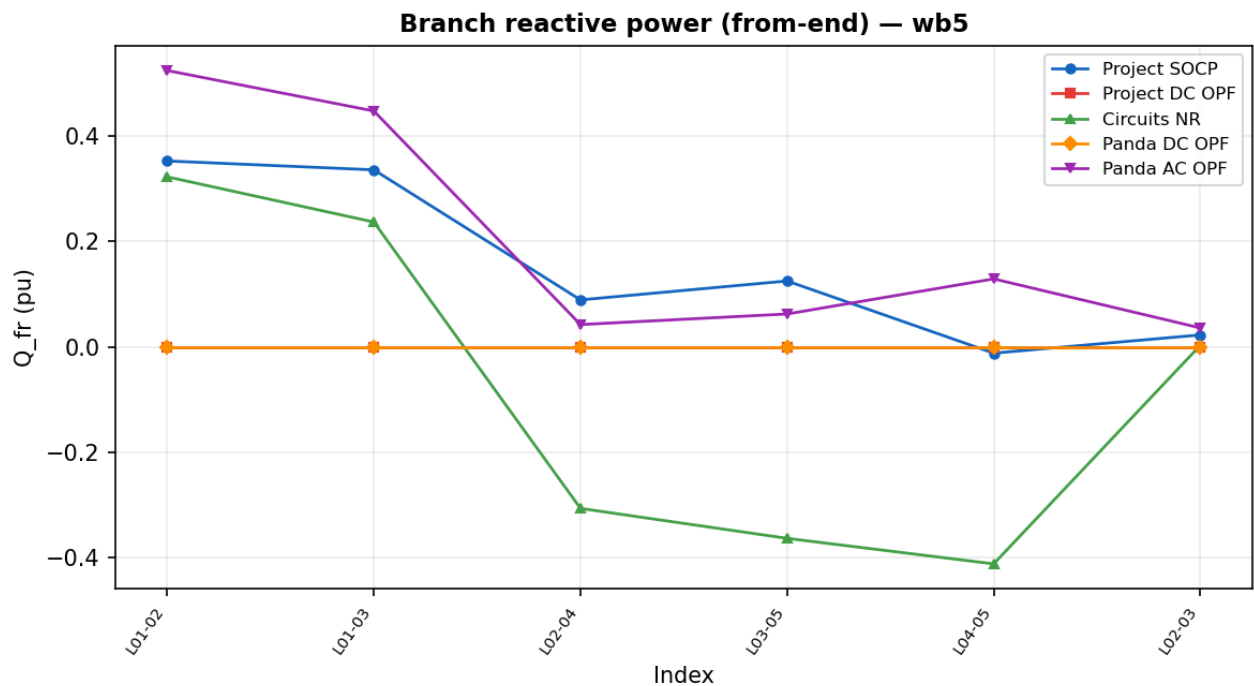


Figure 6. Reactive power Q_{from} (pu). DC solvers return $Q = 0$ by definition. SOCP and AC solvers show substantial reactive flows on L01-02 and L01-03 (slack bus supplies reactive demand). Circuits NR shows negative Q on L02-04 and L03-05 at the physical operating point.